

[Sample Midterm Exam]

Spring 2025: CSE 309: Theory of Automata

- **PLEASE READ THESE INSTRUCTIONS CAREFULLY.**

- Read all questions carefully before you start answering and prioritize which questions to attempt first.
- Please write your name and student ID on this question paper clearly.
- Please do not use pencils or red/green/yellow pens for answers that you want to be graded.
- No queries during the exam.
- If we can't read we can't grade.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
9	10	
10	10	
Total:	100	

– Good Luck –

Question 1

[10 points]

Mark **T** for True and **F** for False.

- (a) _____ Context-free languages are subset of regular languages.
- (b) _____ Every regular language is a context-free language.
- (c) _____ Turing machines can decide regular languages.
- (d) _____ A language is recursive if and only if some NFA accepts it.
- (e) _____ The language $\{0^k w^* : k = 5, w \in \{0, 1\}^*\}$ is a regular language.
- (f) _____ The language $\{w : w \text{ is a palindrome}\}$ is a context-free language.
- (g) _____ Two languages are equal if and only if their complements are equal.
- (h) _____ $\{0^n 1^n : n \geq 0, n \bmod 2 = 0\}$ is a regular language.
- (i) _____ There may be a non-regular language whose complement is regular, but it does not hold for all non-regular languages.
- (j) _____ There is a language A that is recognized by a nondeterministic Turing machine but not by any deterministic Turing machine.

Question 2

[10 points]

Let $\Sigma = \{a, b, c\}$ and let A be a language of strings $w \in \Sigma^*$ such that w starts with an **a**, ends with a **b**, contains at least one **c**, and has length at least 3. Give a regular expression for A and describe a DFA that recognizes A .

Question 3

[10 points]

Use the *pumping lemma* to show that the following language L is not regular:

$$L = \{w\#w : w \in \{0, 1\}^*\}$$

Question 4

[10 points]

Let $\Sigma = \{0, 1, +, =\}$ and A be a language defined as following:

$$A = \{x = y + z : x, y, z \text{ are binary integers, and } x \text{ is the sum of } y \text{ and } z\}.$$

Show that A is not a regular language using the *pumping lemma*.

Question 5

[10 points]

For a language L we define the operation *rotation* as following:

$$\text{rotation}(L) = \{yx : xy \in L\}.$$

Show that if L is a regular language, then $\text{rotation}(L)$ is also a regular language. Furthermore, show that $\text{rotation}(\text{rotation}(L)) = \text{rotation}(L)$ for any language L .

Question 6

[10 points]

Convert the following context-free grammar into an equivalent grammar in Chomsky normal form:

$$\begin{aligned} S &\rightarrow ASA \mid aB \\ A &\rightarrow B \mid S \mid \epsilon \\ B &\rightarrow b \mid CC \\ C &\rightarrow a \mid A \end{aligned}$$

Question 7

[10 points]

Let L_1 and L_2 be two arbitrary Turing-recognizable languages (recursively enumerable languages). Show that:

- (a) The union of L_1 and L_2 is Turing-recognizable.
- (b) The concatenation of L_1 and L_2 is Turing-recognizable.

Question 8

[10 points]

Show that every infinite recursively enumerable language has an infinite recursive subset.

Question 9

[10 points]

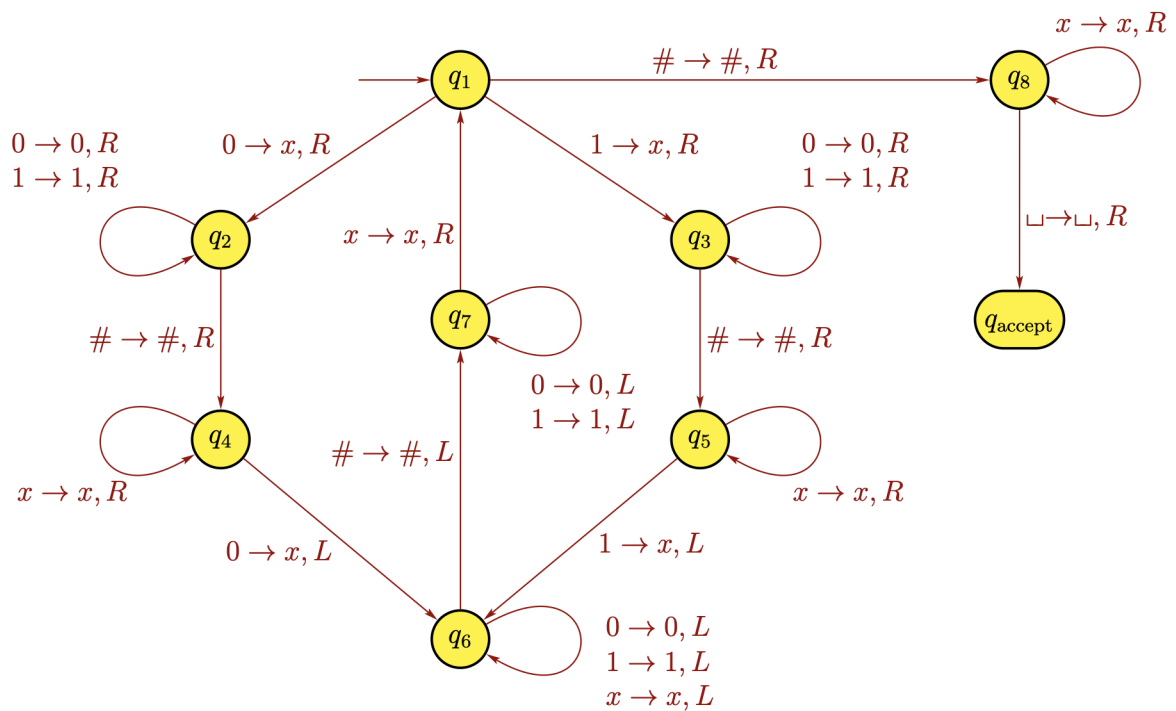
Design an unambiguous context-free grammar for the language:

$$L = \{u\#v : u, v \in \{0, 1\}^*, u \neq v\}.$$

Question 10

[10 points]

Consider the following Turing machine:



In each of the parts below, give the sequence of configurations that M enters when started on the indicated input string.

- (a) **110#01**
- (b) **0#0**